

Adding Integers: The Commutative Property

The Commutative Property of Addition

The operation of addition is **commutative**, meaning numbers can be added in any order without changing the result. For example,

$$\begin{aligned}4 + 3 &= 3 + 4 \\7 &= 7\end{aligned}$$

When adding multiple numbers with different signs, we can use this property to work more efficiently. Instead of working from left to right, simply combine the positive numbers and the negative numbers separately, then add those results.

Example 1. Use the commutative property to find the sum $-6 + 3 + (-4) + 1$.

Explanation. Add the negative numbers -6 and -4 , then add the positive numbers 3 and 1 . Then add the resulting integers to find the answer.

$$\begin{aligned}-6 + 3 + (-4) + 1 &= -10 + 4 \\&= -6\end{aligned}$$

Alt text: Diagram showing how to use commutative property to add negative 6 plus 3 plus negative 4 plus 1. A blue line connects negative 6 and negative 4 with the label "Add Negatives". An orange line connects 3 and 1 with the label "Add Positives". Dashed arrows show these groups simplifying on the right side to negative 10 plus 4. Below this step, the final line shows an equals sign and the simplified result, negative 6.

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Problem 1 Use the commutative property to find each sum.

$$-5 + 2 + (-4) + 8 = 1$$

$$3 + (-1) + 6 + (-15) + 1 = -6$$

$$7 + (-12) + 3 = -2$$

Problem 2 The **Commutative Property of Addition** states that changing the order of the numbers being added does not change the final sum.

Which of the following correctly demonstrates this property?

Multiple Choice:

(a) $-12 + 7 = 12 + (-7)$

(b) $-12 + 7 = 7 + (-12)$

(c) $-12 + 7 = -12 - 7$

(d) $-12 + 7 = 7 + (-12)$ ✓

Find the sum: -5

Problem 3 Use the **Commutative Property of Addition** to find the missing value that makes the addition statement true.

$$450 + (-150) = -150 + 450$$

Find the sum: 300

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Problem 4 A local grassroots organization hosts a community fundraiser to support a local food bank. In the morning, they receive a corporate donation of \$400 (+400). In the afternoon, they spend \$150 on purchasing fresh produce for families in need (−150). According to the **Commutative Property of Addition**, changing the order of these transactions will not alter the organization's final balance. Which of the following equations correctly demonstrates this property by showing that the afternoon transaction was counted first?

Multiple Choice:

(a) $-150 + 400 = -400 + 150$

(b) $-150 + 400 = 400 + (-150)$ ✓

(c) $150 + 400 = 400 + 150$

(d) $150 - 400 = 400 - 150$

Find the organization's final balance: \$ 250

Remark 1. Once you have mastered these addition rules, you will be ready for subtraction. As you will see, if you can add integers, you can subtract them too!